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Lic. in Informatics and Computing Engineering (L.EIC)

Discrete Mathematics

Extra 5 - Proofs in First Order Logic

5.1. Using the natural deduction system for $1^{\underline{st}}$ order logic and without using completeness, show using Fitch notation:

(a) $\forall x(R(x) \land S(x)) \vdash \forall x\, R(x) \land \forall x S(x)$

(b) $\exists x \exists y(P(x,y) \lor P(y,x)) \vdash \exists x \exists y\, P(x,y)$

(c) $\exists x \exists y(H(x,y) \lor H(y,x)), \neg \exists x H(x,x) \vdash \exists x \exists y\, \neg(x = y)$

(d) $\forall x \forall y\, R(x,y) \vdash \forall y \forall x\, R(x,y)$

(e) $\vdash \forall x \forall y \forall u \forall v((x = u \land y = v) \to (P(x,y) \to P(u,v)))$

(f) $\exists y \exists x\, Q(y,x) \vdash \exists x \exists y\, Q(y,x)$

(g) $\exists x \forall y\, P(x,y) \vdash \forall y \exists x\, P(x,y)$

(h) $\forall x(P(x) \to Q(x)) \to (\exists x\, P(x) \to \exists x\, Q(x))$

5.2. Let Σ be a set of formulas of a first order language $\mathcal{L}$ such that x does not occur free in any formula of Σ. Show that,

(a) if $\Sigma \cup \{\varphi\} \vdash \psi$ and x does not occur free in ψ, then $\Sigma \cup \{\exists x \varphi\} \vdash \psi$;

(b) if $\Sigma \cup \{\varphi\} \vdash \psi$, then $\Sigma \cup \{\exists x \varphi\} \vdash \exists x \psi$.

Note: In both cases, also assume that x does not occur free in φ. The stronger without this assumption requires an additional substitution lemma.

5.3. Let ψ be a proposition of a first-order language $\mathcal{L}$. List the following statements, indicating if they are equivalent or if only one implies the other **justifying**.

(a)
- there is a structure $\mathcal{A}$ of $\mathcal{L}$ and a interpretation s of variables in $\mathcal{A}$ such that $\mathcal{A} \not\models_s \neg\psi$;
- ψ is a valid formula.

(b)
- $\vdash \psi$
- $\neg\psi$ is false in all structures of $\mathcal{L}$.

(c)
- $\models \psi$
- there are two structures $\mathcal{A}$ and $\mathcal{B}$ from $\mathcal{L}$ such that $\mathcal{A} \models \neg\psi$ and $\mathcal{B} \not\models \neg\psi$.

	Introduction	**Elimination**
$\wedge$	φ $\vdots$ ψ $\vdots$ $\varphi \wedge \psi$ $\wedge$I	$\vdots$ $\varphi \wedge \psi$ $\vdots$ φ $\wedge$E $\vdots$ $\varphi \wedge \psi$ $\vdots$ ψ $\wedge$E
$\vee$	$\vdots$ φ $\vdots$ $\varphi \vee \psi$ $\vee$I $\vdots$ φ $\vdots$ $\psi \vee \varphi$ $\vee$I	$\varphi \vee \psi$ $\vdots$ φ / $\vdots$ / γ ψ / $\vdots$ / γ γ $\vee$E
$\neg$	φ / $\vdots$ / $\mathbf{F}$ $\neg\varphi$ $\neg$I	$\vdots$ $\neg\neg\varphi$ $\vdots$ φ $\neg$E
$\mathbf{F}$	φ $\vdots$ $\neg\varphi$ $\vdots$ $\mathbf{F}$ **FI**	$\vdots$ $\mathbf{F}$ $\vdots$ φ **FE**
$\rightarrow$	φ / $\vdots$ / ψ $\varphi \rightarrow \psi$ $\rightarrow$I	φ $\vdots$ $\varphi \rightarrow \psi$ $\vdots$ ψ $\rightarrow$E
$=$	$\vdots$ $t = t$ $=_{\mathbf{I}}$	$t_1 = t_2$ $\vdots$ $\varphi[x \mapsto t_1]$ $\vdots$ $\varphi[x \mapsto t_2]$ $=_{\mathbf{E}}$ and x is replaceable by t_1 and by t_2 in φ
$\forall$	ν / $\vdots$ / $\varphi[x \mapsto \nu]$ $\forall x\ \varphi$ $\forall_{\mathbf{I}}$ where ν is a new variable (does not occur before)	$\forall x\ \varphi$ $\vdots$ $\varphi[x \mapsto t]$ $\forall_{\mathbf{E}}$ where x is replaceable by t in φ
$\exists$	$\varphi[x \mapsto t]$ $\vdots$ $\exists x\ \varphi$ $\exists_{\mathbf{I}}$ where x is replaceable by t in φ	$\exists x\ \varphi$ ν / $\varphi[x \mapsto \nu]$ / $\vdots$ / ψ ψ $\exists_{\mathbf{E}}$ where ν is a new variable that does not occur before or in ψ